Retail Rag

Course Name: Gen AI

Institution Name: Medicaps University – Datagami Skill Based Course

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1. Problem Statement & Objectives

1.1 Problem Statement

In today's retail and business environments, organizations deal with a massive amount of unstructured data such as invoices, reports, manuals, and customer-related documents. Extracting meaningful insights from these documents is challenging and time-consuming.

Traditional keyword-based search systems are inefficient because they:

- Fail to understand context
- Provide irrelevant or incomplete results
- Cannot handle natural language queries effectively

Users often need to manually search through large documents, leading to:

- Reduced productivity
- Increased effort
- Poor decision-making due to lack of insights

There is a need for an intelligent system that can:

- Understand natural language queries
- Retrieve relevant document information
- Generate accurate, context-aware responses

1.2 Project Objectives

- **Enable natural language interaction with documents**

Develop a system that allows users to query documents using simple natural language, making document exploration easier without requiring technical knowledge or complex search techniques.

- **Implement RAG for accurate responses**

Use Retrieval-Augmented Generation to combine document retrieval with AI generation, ensuring responses are accurate, relevant, and based on actual document content rather than assumptions.

- **Process and embed document text**

Extract text from documents like PDF and DOCX, divide it into smaller chunks, and convert them into embeddings for efficient semantic understanding and retrieval.

- **Perform semantic search**

Implement similarity-based search using embeddings to find the most relevant document content, improving search quality compared to traditional keyword-based methods.

- **Generate context-aware answers**

Use Large Language Models to generate meaningful answers by considering retrieved document context, ensuring responses are relevant, coherent, and aligned with user queries.

- **Improve efficiency and scalability**

Design a system that reduces manual effort, provides faster responses, supports multiple users and documents, and ensures scalability for real-world applications.

1.3 Scope of the Project

The scope of this project focuses on developing an AI-powered document interaction system that enables users to query and extract insights from unstructured documents using natural language.

Included Scope:

- Accepting user-uploaded documents such as PDF and DOCX files
- Extracting and processing text from uploaded documents
- Splitting text into chunks and generating embeddings
- Storing embeddings using vector databases (FAISS / ChromaDB)
- Performing semantic similarity search to retrieve relevant content
- Generating context-aware responses using Large Language Models (LLMs)
- Providing a chat-based user interface for interaction
- Implementing REST APIs for communication between frontend and backend

Limitations:

- The system supports only text-based document formats
- Output accuracy depends on the quality and clarity of uploaded documents
- May not handle highly complex or ambiguous queries perfectly
- Requires external API access for LLM functionality
- Does not include advanced features like multi-language support or voice input

2. Proposed Solution

2.1 Key Features

The proposed system incorporates several advanced features to enable efficient and intelligent document interaction and information retrieval:

1. Document Upload and Processing
   Users can upload documents such as PDF and DOCX through the frontend interface. The system processes these documents by extracting text for further analysis.

2. Text Chunking and Embedding Generation
   The extracted text is divided into smaller chunks and converted into embeddings, enabling efficient semantic understanding and storage of document content.

3. Vector-Based Storage and Retrieval
   Embeddings are stored in a vector database (FAISS / ChromaDB), allowing fast and efficient similarity-based retrieval of relevant document information.

4. Semantic Search Capability
   The system performs similarity search using embeddings to identify the most relevant document chunks based on the user's query instead of relying on keyword matching.

5. AI-Powered Response Generation
   A Large Language Model (LLM) is used to generate accurate and meaningful answers by combining user queries with retrieved document context.

6. Context-Aware Answer Generation
   The system ensures that responses are generated based on actual document content, improving relevance and reducing incorrect or unrelated outputs.

7. Reduced Hallucination
   By using a Retrieval-Augmented Generation (RAG) approach, the system minimizes hallucination and ensures responses are grounded in real data.

8. Fast and Scalable Architecture
   The system is designed with a modular and scalable architecture, enabling efficient processing, low latency responses, and support for multiple users and documents.

2.2 Overall Architecture / Workflow

The system architecture is designed as a multi-layered pipeline that ensures efficient document processing and accurate response generation.

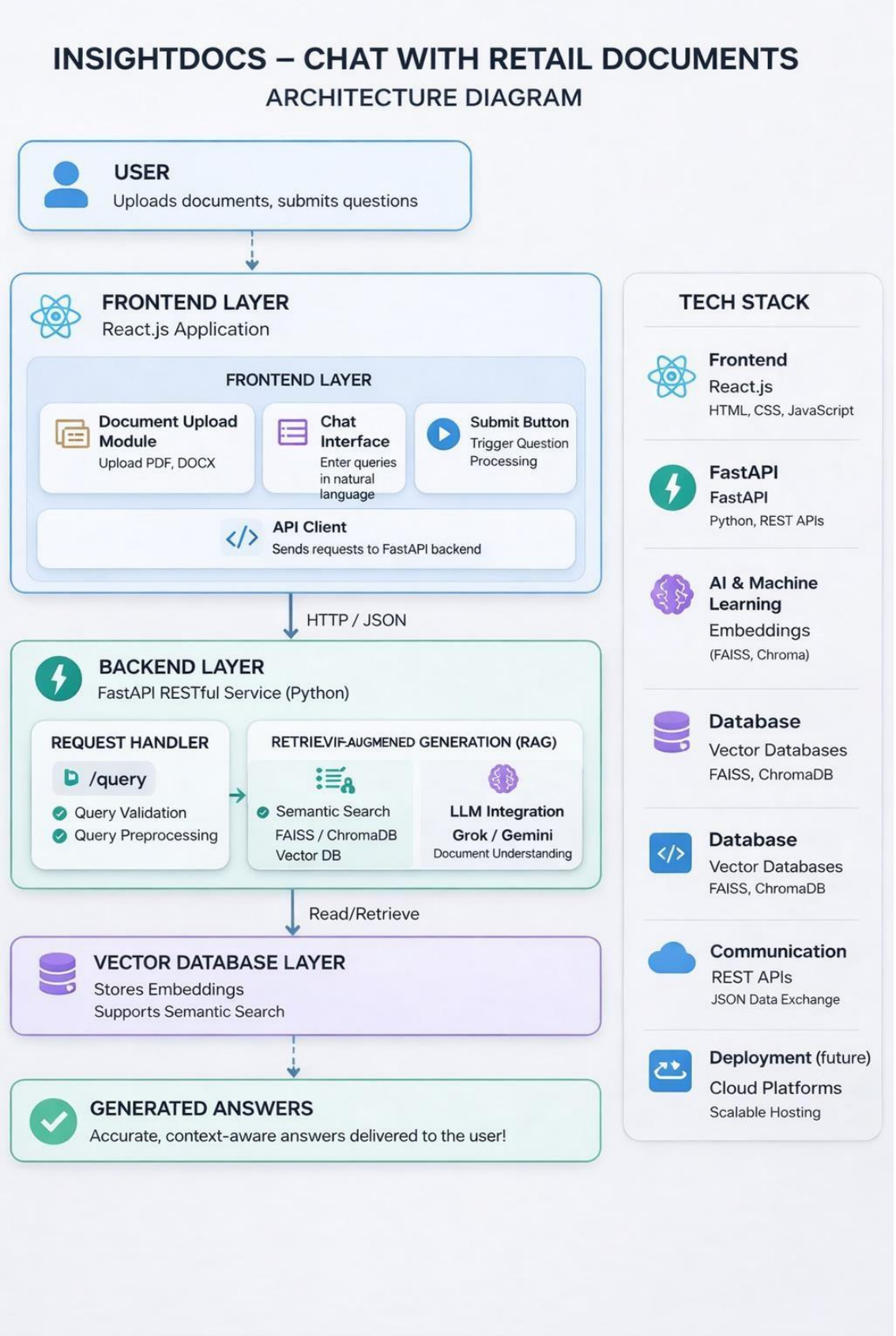
Phase 1: Document Processing Pipeline
• The user uploads documents such as PDF or DOCX through the frontend interface.
• The backend (FastAPI) receives the document and initiates processing.
• Text is extracted from the uploaded document.
• The extracted text is divided into smaller chunks for better processing.
• Each text chunk is converted into embeddings using an embedding model.
• The generated embeddings are stored in a vector database for efficient retrieval.

Phase 2: Query Processing Pipeline
• The user submits a query through the chat interface.
• The query is converted into an embedding representation.
• A similarity search is performed in the vector database to find relevant document chunks.
• Retrieved document context is passed to the Large Language Model (LLM).
• The LLM generates a context-aware and accurate response.
• The final answer is sent back to the user through the frontend interface.

This workflow ensures efficient document understanding, accurate information retrieval, and a seamless user experience.

Key Workflow Characteristics:

- Two-stage pipeline: Document processing and query processing ensure efficient workflow
- Structured processing: Uses chunking and embeddings for consistent and accurate results
- Context-aware generation: Responses are generated based on user query and retrieved document context
- Efficient processing: Optimized backend and vector search ensure fast and reliable response


INSIGHTDOCS – CHAT WITH RETAIL DOCUMENTS
ARCHITECTURE DIAGRAM
USER
Uploads documents, submits questions
FRONTEND LAYER
React.js Application
FRONTEND LAYER
Document Upload Module
Upload PDF, DOCX
Chat Interface
Enter queries in natural language
Submit Button
Trigger Question Processing
API Client
Sends requests to FastAPI backend
HTTP / JSON
BACKEND LAYER
FastAPI RESTful Service (Python)
REQUEST HANDLER
/query
Query Validation
Query Preprocessing
RETRIEVIF-AUGMENED GENERATION (RAG)
Semantic Search
FAISS / ChromaDB
Vector DB
LLM Integration
Grok / Gemini
Document Understanding
Read/Retrieve
VECTOR DATABASE LAYER
Stores Embeddings
Supports Semantic Search
GENERATED ANSWERS
Accurate, context-aware answers delivered to the user!
TECH STACK
Frontend
React.js
HTML, CSS, JavaScript
FastAPI
FastAPI
Python, REST APIs
AI & Machine Learning
Embeddings
(FAISS, Chroma)
Database
Vector Databases
FAISS, ChromaDB
Database
Vector Databases
FAISS, ChromaDB
Communication
REST APIs
JSON Data Exchange
Deployment (future)
Cloud Platforms
Scalable Hosting

2.3 Tools & Technologies Used

The system integrates multiple technologies across different layers to enable efficient document processing and intelligent response generation:

Frontend Technologies
• React.js for building a dynamic and interactive user interface
• HTML, CSS for designing responsive UI components
• Enables real-time document upload and chat-based interaction

Backend Technologies
• FastAPI for handling HTTP requests and implementing RESTful APIs
• Python as the primary programming language for backend logic and processing

AI and Machine Learning Components
• Embedding models for converting text into vector representations
• Large Language Models (LLMs) for generating context-aware responses
• Retrieval-Augmented Generation (RAG) for improving accuracy and reducing hallucination

Database
• Vector databases (FAISS / ChromaDB) for storing embeddings
• Supports efficient similarity search and fast information retrieval

Supporting Tools
• REST APIs for communication between frontend and backend
• JSON for structured data exchange between system components

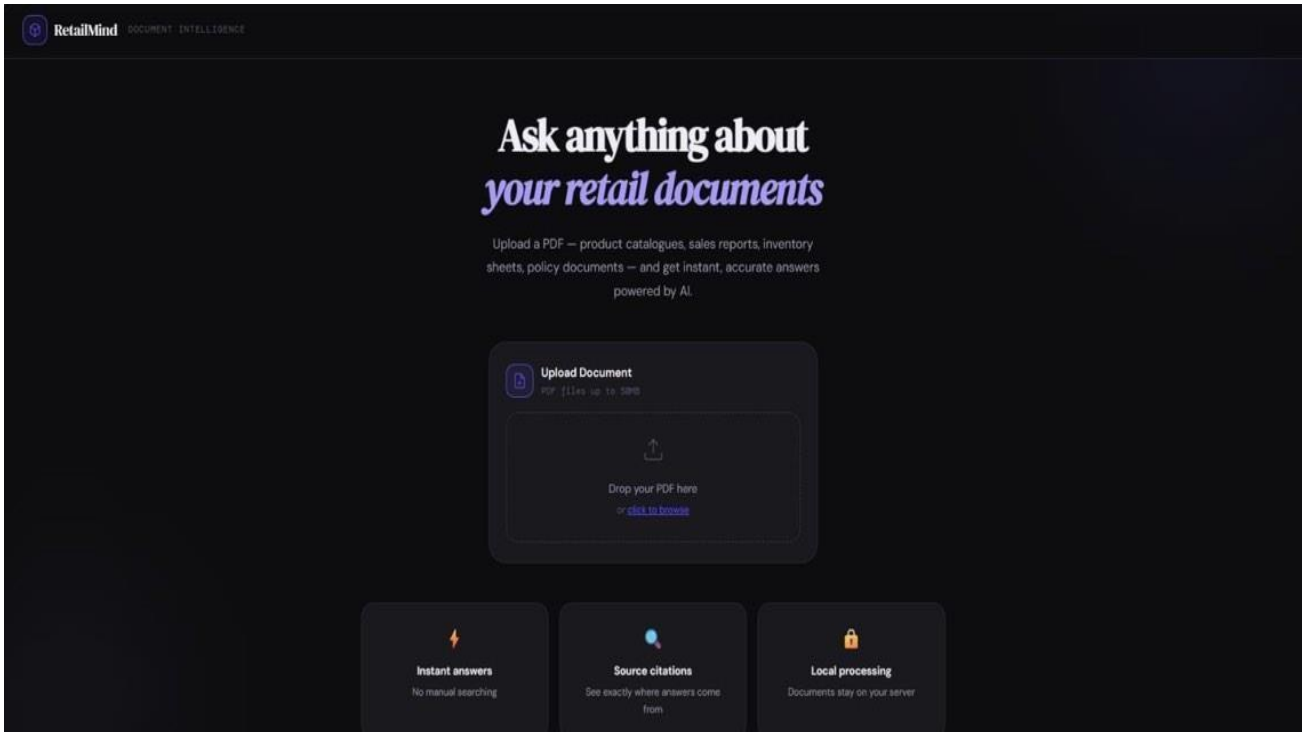
3. Results & Output

3.1 Screenshots / Outputs

The system provides a user-friendly interface that enables seamless document interaction and information retrieval:

• A document upload module where users can upload files such as PDF and DOCX
• A chat interface where users can enter natural language queries
• A submit button to process queries and retrieve relevant information
• An output display area where AI-generated, context-aware responses are presented

The generated outputs include accurate and well-structured answers based on document content. These outputs demonstrate the system's ability to efficiently retrieve and present relevant information.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your retail documents
Upload a PDF — product catalogues, sales reports, inventory sheets, policy documents — and get instant, accurate answers powered by AI.
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See exactly where answers come from
Local processing
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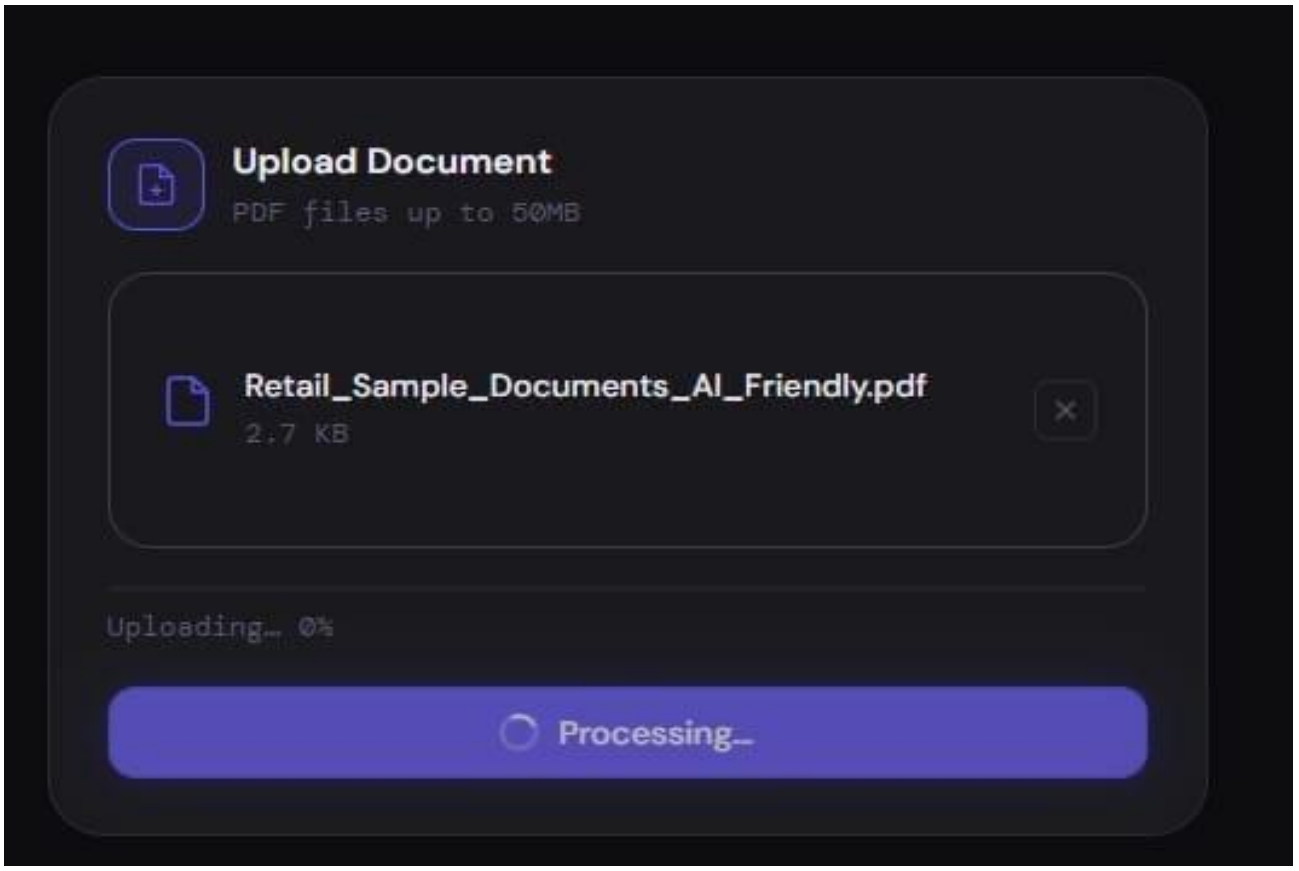
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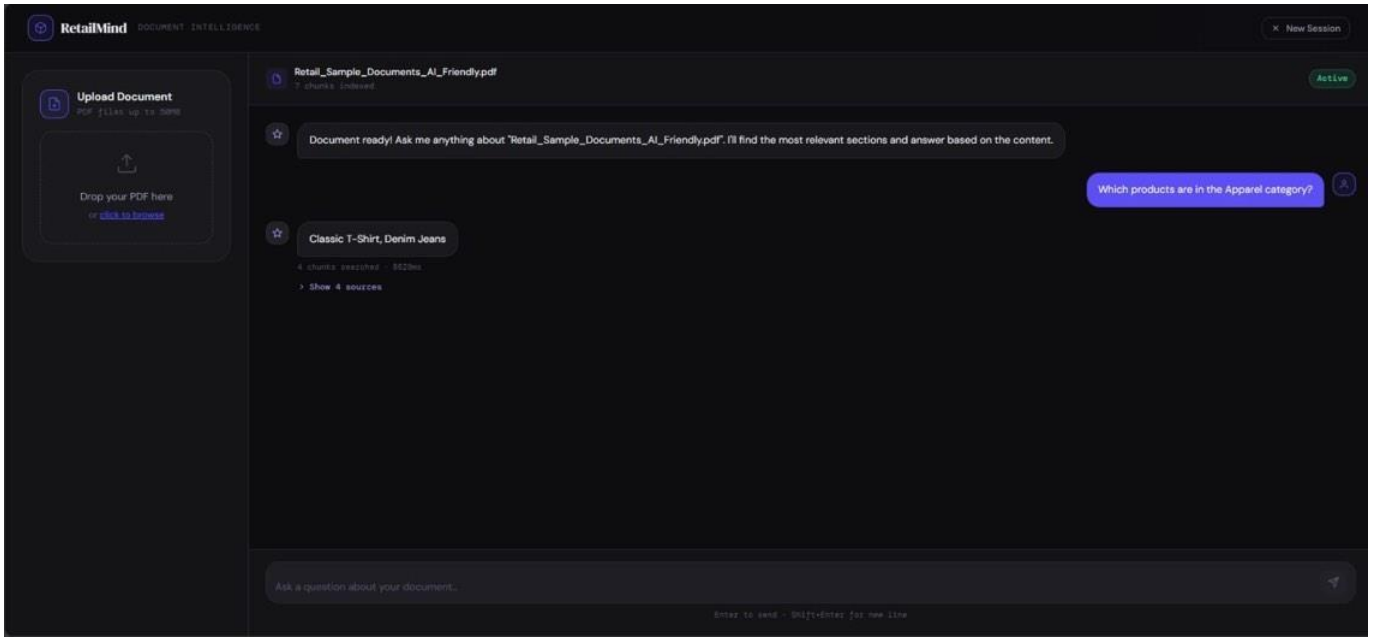

3.2 Key Outcomes

The implementation of the system resulted in the following outcomes:

• Successful development of an AI-powered document question-answering system

• Accurate and context-aware responses generated using Large Language Models (LLMs)

• Improved information retrieval through semantic search and embeddings

• Significant reduction in manual effort required to search and analyze documents

• Seamless integration between frontend, backend, and AI components

• Enhanced user experience through simple chat-based interaction

• Efficient handling of multiple documents and user queries

4. Conclusion

The InsightDocs system successfully demonstrates the application of Generative AI in transforming document analysis and information retrieval.

The system effectively utilizes Retrieval-Augmented Generation (RAG), embeddings, and Large Language Models (LLMs) to generate accurate, context-aware responses from unstructured documents. It simplifies the process of extracting information by allowing users to interact with documents using natural language, significantly reducing manual effort and time.

By leveraging a modular and scalable architecture, the system ensures efficient performance, fast retrieval, and easy integration with future technologies. It also enhances user experience through a simple chat-based interface without requiring technical expertise.

This project highlights the potential of AI-driven solutions in converting traditional document search into an intelligent, automated, and highly efficient conversational system.

5. Future Scope & Enhancements

The InsightDocs system can be further improved and extended in several ways:

Technical Enhancements

• Support for additional file formats such as Excel, images (with OCR), and scanned documents
• Integration of multilingual support for processing documents in different languages
• Implementation of voice-based query input for hands-free interaction

AI Enhancements

• Fine-tuning of Large Language Models for domain-specific document understanding
• Improved retrieval techniques for better accuracy and relevance
• Integration of summarization and advanced analytics features

System Enhancements

• User authentication and profile management
• Storage of user query history and document management system
• Deployment on cloud platforms for better scalability and availability

Real-World Applications

• Integration with enterprise document management systems
• Use in industries like retail, finance, healthcare, and education
• Deployment as a SaaS-based document intelligence platform